

# Rodrigo Roque-Hernandez

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Applied AI & Data Systems Builder — Human-Centered Product Design

## EDUCATION

### University of North Carolina at Chapel Hill

Chapel Hill, NC

Bachelor of Science in Information Science & Data Science

August 2023 – May 2027

Awards & Activities:

HSF Scholarship Award, SHPE, CAMpFIRE Scholarship

## TECHNICAL EXPERIENCE

### Director of Web Development / Technical Lead

Aug 2025 – Present

Multi-Org Platforms

– React, Firebase, HTML, CSS, JavaScript, Figma

- Led cross-functional teams of developers and designers, driving frontend, backend, and UX execution while coordinating technical direction across multi-organization platforms
- Designed UX workflows and modular interface systems supporting event registration, content publishing, and hackathon platforms
- Integrated analytics tracking and platform optimization strategies, increasing user engagement and session duration by 23%

### ecDNA Research Assistant

Oct. 2024 – May 2025

Brunk Cytogenomics Lab

– Python, PyTorch, OpenCV, Pandas

- Developed deep learning computer vision pipelines to detect and localize extra-chromosomal DNA (ecDNA) in FISH microscopy images, improving analysis efficiency by 25%
- Built automated preprocessing workflows including normalization, augmentation, and coordinate mapping to scale biomedical imaging datasets
- Quantified model performance and structural patterns in cancer imaging data, reducing manual review time by 12%

### Computational Research Assistant

Sep 2024 – May 2025

Baker Lab

– Python, Unix, Jupyter Notebooks, Unipressed

- Automated ingestion and processing of 200+ protein records (UniProt, PDB) to enhance AlphaFold-based structural analysis pipelines
- Built data visualization workflows supporting biological inference and experimental decision-making
- Contributed 50K+ lines of code to an established codebase via Git
- Optimized reproducible data workflows for large-scale biological datasets

## ENGINEERING & PRODUCT PROJECTS

### Diabetes Readmission Prediction Model | Python, scikit-learn,

- Built Decision Tree and Random Forest models to predict 30-day hospital readmission risk using structured clinical datasets
- Designed preprocessing pipelines using ColumnTransformer to handle missing values and categorical features
- Evaluated performance with ROC curves and classification metrics (Test AUC = 0.656)

### SolHacks Platform | Figma, React, UX Architecture

- Designed full mobile and desktop interface system including onboarding flows, schedules, and team management modules
- Built reusable component structures bridging product design and frontend engineering workflows
- Applied usability-driven interaction design to support a 150+ attendee hackathon platform
- Collaborated with organizing leadership and engineering teams to align platform design with event strategy and participant experience goals

## TECHNICAL SKILLS

**Languages:** Python, R, JavaScript, SQL, HTML/CSS

**Frameworks:** React, WordPress

**Developer Tools:** Git, VS Code, JupyterLab, Unix, Google Colab

**Libraries:** PyTorch, scikit-learn, pandas, NumPy, Matplotlib, seaborn

**Design & Product:** Figma, UX Prototyping, Component Systems